Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following has the maximum number of resonating structures?

(A) Benzene

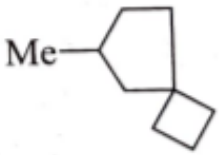
(B) Anthracene

(C) Naphthalene

(D) Phenanthrene

Q.2

The systematic nomenclature of the following spirocompound is:



(A) $2-$Methyl spiro $[3.4]$ octane

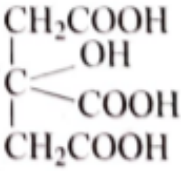
(B) $3-$Methyl spiro $[3.4]$ octane

(C) $6-$Methyl spiro $[3.4]$ octane

(D) $7-$Methyl spiro $[3.4]$ octane

Q.3

The IUPAC name of



(A) $3-$Carboxy$-3-$hydroxypentanedicarboxylic acid

(B) $2-$Hydroxypropane$-1,2,3-$tricarboxylic acid

(C) $2-$Hydroxypropane$-1,2,3-$trioic acid

(D) $3-$Hydroxypropane$-1,2,3-$tricarboxylic acid

Q.4

Which of the following is a FALSE statement?

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(A) Carius tube is used in the estimation of sulphur in an organic compound.

(B) Phosphoric acid produced on oxidation of phosphorus present in an organic compound is precipitated as $Mg_2P_2O_7$ by adding magnesia mixture.

(C) Carius method is used for the estimation of nitrogen in an organic compound.

(D) Kjeldahl's method is used for the estimation of nitrogen in an organic compound.

Q.5

For the separation of two miscible liquids which method is used?

(A) Chromatography

(B) Fractional column

(C) Fractional distillation

(D) Separating funnel

Q.6

The kind of delocalization involving sigma bond orbitals is called

(A) Inductive effect

(B) Electromeric effect

(C) Hyperconjugation effect

(D) Mesomeric effect

Q.7

In the estimation of nitrogen by Kjeldahl's method, 2.8 g of an organic compound required 20 millimole of H_2SO_4 for the complete neutralisation of NH_3 gas evolved. The percentage of nitrogen in the sample is:

(A) 20%

(B) 10%

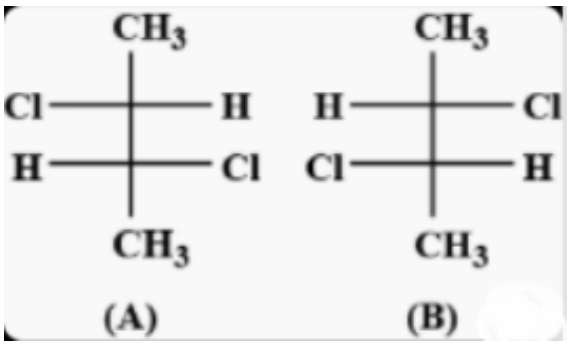
(C) 40%

(D) 30%

Q.8

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If optical rotation produced by the compound (A) is $+52^{\circ}$, the one produced by compound (B) is :



(A) -52°

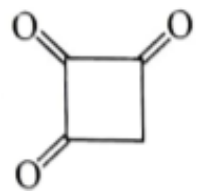
(B) $+52^{\circ}$

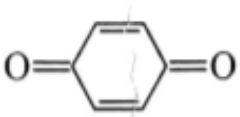
(C) 0°

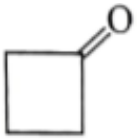
(D) Unpredictable

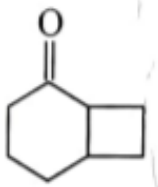
Q.9

Tautomerism is not exhibited by:

(A) 

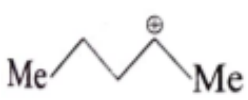
(B) 

(C) 

(D) 

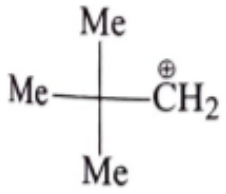
Q.10

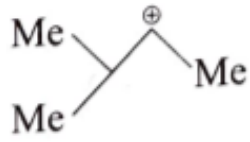
Which of the following carbocations is most stable?

(A) 

(B) 

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(C) 

(D) 

Q.11

The decreasing order of $-\mathrm{I}$ effect of the following is:

I. COOH II. F III. OR IV. NH_2 V. OH VI. Ph-

(A) $\mathrm{I} > \mathrm{II} > \mathrm{III} > \mathrm{IV} > \mathrm{V} > \mathrm{VI}$

(B) $\mathrm{II} > \mathrm{I} > \mathrm{III} > \mathrm{IV} > \mathrm{V} > \mathrm{VI}$

(C) $\mathrm{I} > \mathrm{II} > \mathrm{V} > \mathrm{III} > \mathrm{IV} > \mathrm{VI}$

(D) $\mathrm{II} > \mathrm{I} > \mathrm{V} > \mathrm{II} > \mathrm{IV} > \mathrm{VI}$

Q.12

Select correct statement about Inductive effect?

(A) Inductive effect transfers electrons from one carbon atom to another.

(B) Inductive effect is the polarisation of bond electrons.

(C) Inductive effect creates net charge in the molecule.

(D) Inductive effect is distance independent.

Q.13

Give the decreasing order of hyperconjugative effect of R in $\mathrm{R} - \mathrm{CH} = \mathrm{CH}_2$, where R is:

I. $\mathrm{Me}-$

II. $\mathrm{Et}-$

III. $\mathrm{Me_2CH}-$

IV. $\mathrm{Me_3C}-$

(A) $\mathrm{I} > \mathrm{II} > \mathrm{III} > \mathrm{IV}$

(B) $\mathrm{IV} > \mathrm{III} > \mathrm{II} > \mathrm{I}$

(C) $\mathrm{II} > \mathrm{I} > \mathrm{II} > \mathrm{IV}$

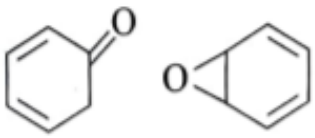
(D) $\mathrm{IV} > \mathrm{III} > \mathrm{I} > \mathrm{II}$

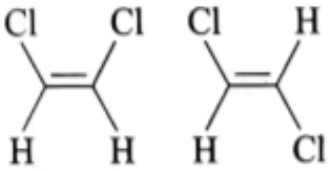
Q.14

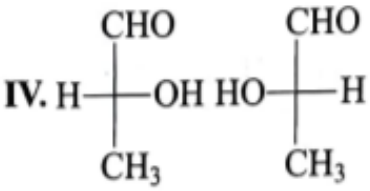
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The relationship among the following pairs of isomers is:

I. 

II. 

III. 

IV. 

I	A : Constitutional
II	B : Structural
III	C : Geometrical
IV	D : Optical

(A) I – A, II – B, III – B, IV – D

(B) I – A, II – A, III – C, IV – D

(C) I – B, II – A, III – B, IV – D

(D) I – B, II – B, III – A, IV – B

Q.15

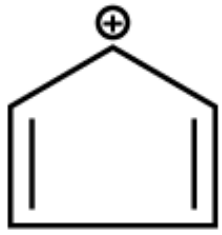
Which of the following is anti-aromatic in nature?

(A)

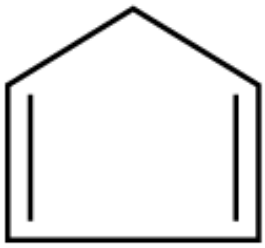


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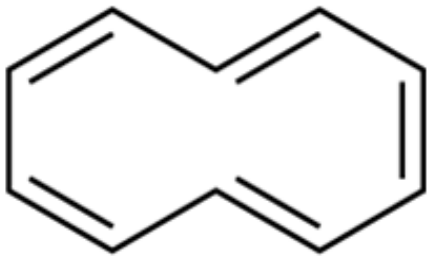
(B)



(C)



(D)



Q.16

Which of the following has highest resonance energy?

(A)



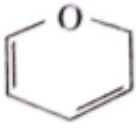
(B)



(C)



(D)



Q.17

How many geometrical isomers are possible for the following compounds:

3—Bromo—5—iodo methylcyclohexane

(A) 6

(B) 7

(C) 4

(D) 5

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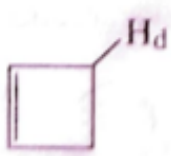
Q.18

Which of the following H will have lower pK_a value in the given compound?

(A) 

(B) 

(C) 

(D) 

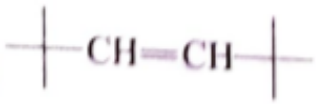
Q.19

In which of the following alkene hyperconjugation does not take place?

(A) 

(B) 

(C) 

(D) 

Q.20

Among the following, the paramagnetic species is:

(A) free radical

(B) carbocation

(C) carbanion

(D) All of these

Q.21

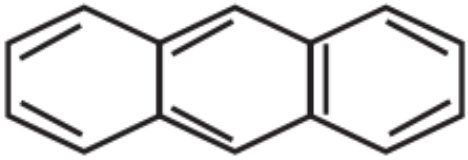


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;
How many 2° carbon present in above compound?

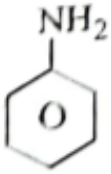
Q.22

The number of resonance structures for Anthracene are

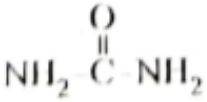


Q.23

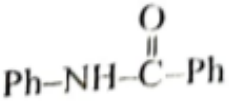
Amongst the following, the total number of compound which does not give Lassaigne's test for nitrogen.
I.



II. H_2N-CH_2-COOH
III. NH_2-NH_2
IV. $Ph-N=N-Ph$
V. NH_2OH
VI.



VII.



VIII. $PhNO_2$
IX. CH_3CHO
X. CH_3CN
XI.



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Q.24

How many of these compounds are more acidic than phenol here.
Formic acid, Benzoic acid, Picric acid, Ethanol, Water, Ortho-nitrophenol, Ortho-cresol, para-nitrophenol, para-cresol

Q.25

The number of geometrical isomers in $CH_3CH = N - OH$ is:

Q.26

Among the following, the number of aromatic compoud(s) is:
(A)



(B)



(C)



(D)



(E)



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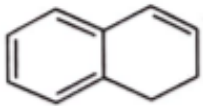
(F)



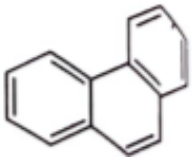
(G)



(H)



(I)

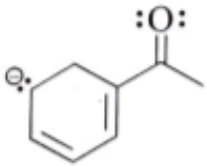


Q.27

How many structural isomers of molecular formula C_3H_6BrCl?

Q.28

How many total resonance structures can be drawn for the following anion (include those without separation of charge)?



Q.29

Number of alpha-hydrogen in given compound is?



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Q.30

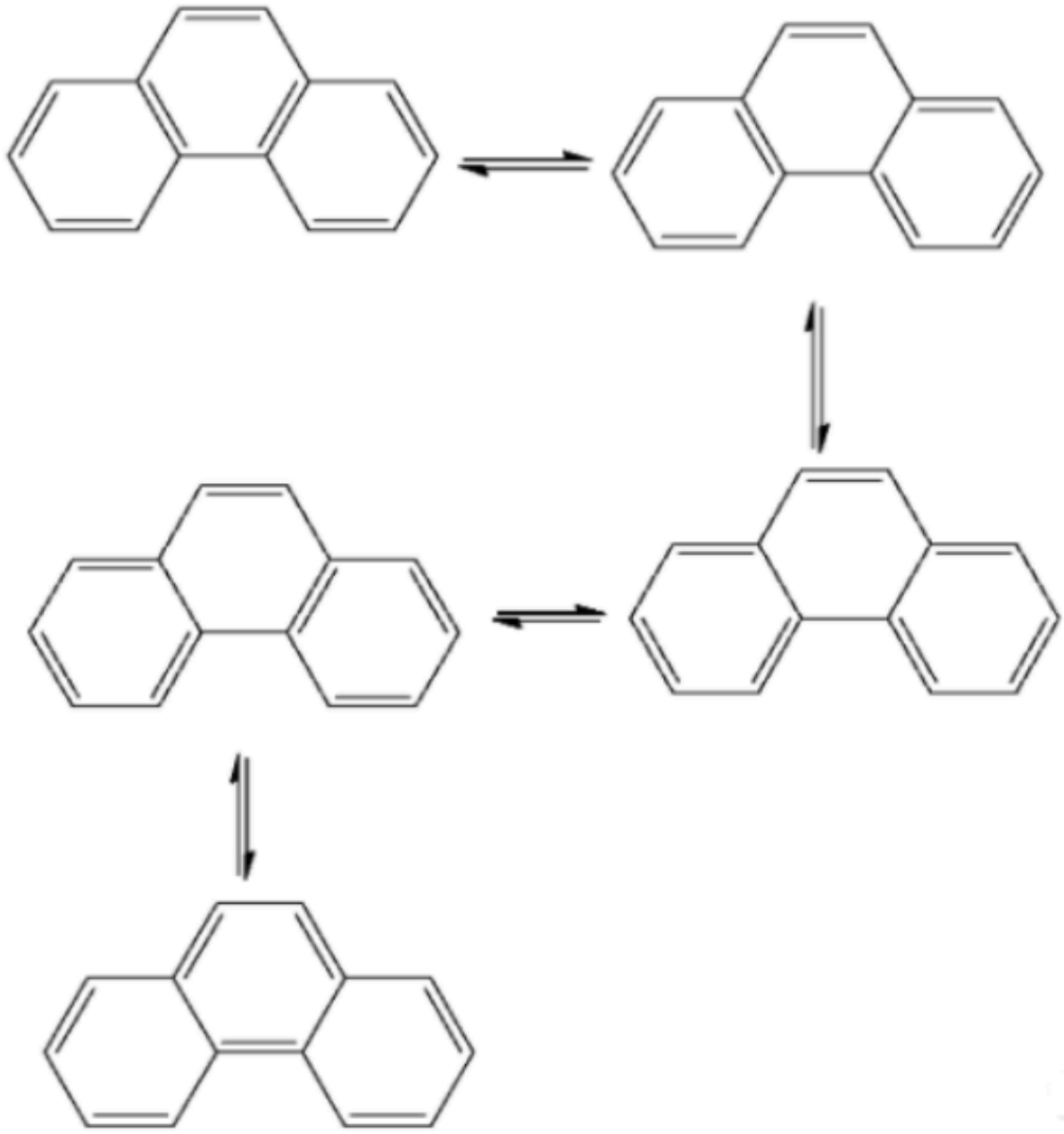
How many optically active stereoisomers are possible for butane$-2,3-$diol?

Answers & Solutions

Q.1 Answer:

Phenanthrene

Solution:



- Therefore Phenanthrene shows five resonating structures.

- Among the given options Phenanthrene shows the highest number of resonating structures.

Q.2 Answer:

$6-$Methyl spiro $[3.4]$ octane

Solution:

$6-$**Methyl spiro** $[3.4]$ **octane**

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Q.3 Answer:

$2-$Hydroxypropane$-1,2,3-$tricarboxylic acid

Solution:

$2-$Hydroxypropane$-1,2,3-$tricarboxylic acid

Q.4 Answer:

Carius method is used for the estimation of nitrogen in an organic compound.

Solution:

Carius method is used in the estimation of halogen in organic compounds.

Q.5 Answer:

Fractional distillation

Solution:

Fractional distillation is the most effective method of separating mixtures of miscible volatile liquids. It is used when the difference in the boiling points of liquids to be separated is less than $20\ \mathrm{K}$.
Simple distillation is used to purify liquid containing non-volatile impurities. It is used when the difference in the boiling points of liquids to be separated is between $30\ \mathrm{K}$ to $50\ \mathrm{K}$. It is used for liquids that do not decompose on heating.
Chromatography is used to separate a mixture having a different affinity for stationary phase and mobile phase.
Filtration is used to separate a solid from a liquid.

Q.6 Answer:

Hyperconjugation effect

Solution:

The kind of delocalization involving sigma bond orbitals is called hyperconjugation.
Hyperconjugation is also known as $\sigma - \pi$ conjugation. It is observed in alkyl groups with at least one hydrogen atom on the alpha carbon atom attached to an unsaturated carbon atom.

Q.7 Answer:

20%

Solution:

Percentage of $\mathrm{N} =$

$$\frac{1.4 \times \text{mEq. of } \mathrm{H_2SO_4} \text{ used to neutralise } \mathrm{NH_3}}{\text{Weight of compound}}$$

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millimoles of $H_2SO_4 = 20$
milliequivalent of $H_4SO_4 = 20 \times 2 = 40$
Percentage of N in the sample $= \frac{1.4 \times 40}{2.8} = 20\%$

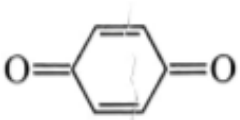
Q.8 Answer:

-52°

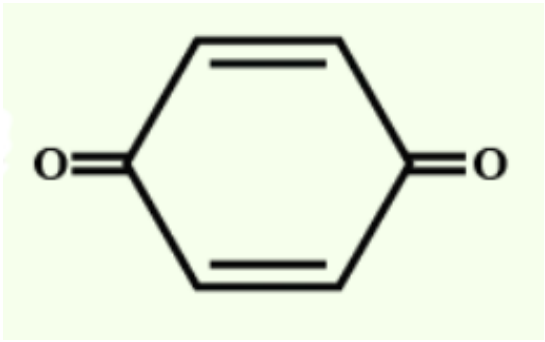
Solution:

Since (A) and (B) are enantiomers, so specific rotation of B is -52° (because enantiomers have equal and opposite specific rotation

Q.9 Answer:



Solution:



can't exhibit tautomerism. By transferring of $\alpha - H$ to oxygen will result in formation cummulated double bond favoring bond angle of 180° making ring very unstable. So, tautomerism will not occur here.

Q.10 Answer:



Solution:

The more number of alpha hydrogens, the more is the stability. since option (a) have more number of alpha hydrogens, it is the most stable

Q.11 Answer:

$I > II > V > III > IV > VI$

Solution:

The decreasing order of -I effect is
I. $-COOH >$ **II.** $-F >$ **V.** $- OH >$ **III.** $-OR >$ **IV.** $- NH_2 >$ **VI. Ph-** @IITJEE_Advanced

The electron withdrawing nature of the atoms/species is called -I Effect. The correct order is-

$I > II > V > III > IV > VI$

Q.12 Answer:

Inductive effect is the polarisation of bond electrons.

Solution:

In chemistry, the inductive effect is an effect regarding the transmission of unequal sharing of the bonding electron through a chain of atoms in a molecule, leading to a permanent dipole in a bond. It is present in a σ (sigma) bond, unlike the electromeric effect which is present in a π (pi) bond.

Q.13 Answer:

$I > II > III > IV$

Solution:

The more number of alpha hydrogens, the more is the hyperconjugative effect.

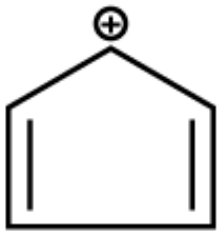
Q.14 Answer:

$I - A, II - A, III - C, IV - D$

Solution:

$I - A, II - A, III - C, IV - D$

Q.15 Answer:



Solution:

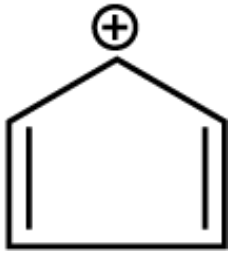
$(A) \Rightarrow$ **Non-aromatic**

(B) $\Rightarrow$ **anti-aromatic (** $4\pi \bar{e}'s$ **conjugation)**

$(C) \Rightarrow$ **non-aromatic (** $4\pi \overrightarrow{e}$ **'s not in resonance)**

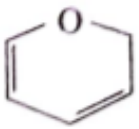
(D) $\Rightarrow [10]$**-Annulene, non-aromatic.**

So, (B) is anti-aromatic.

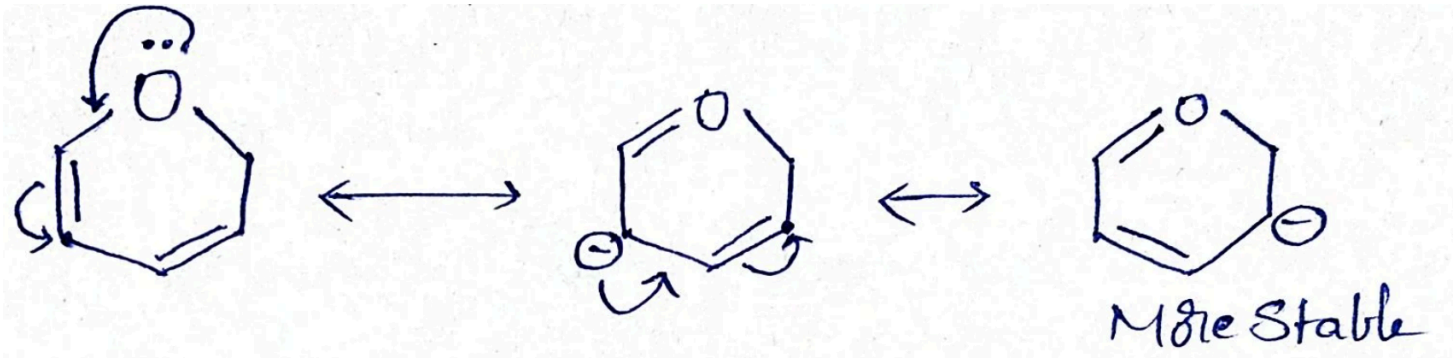
(b) $\Rightarrow$ 

Q.16 Answer:

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Solution:



Linear Resonance

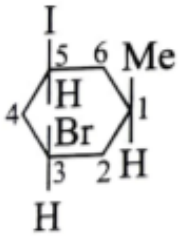
Q.17 Answer:

4

Solution:

Four geometrical isomers:

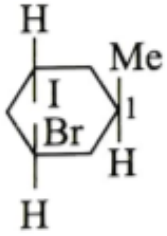
(I)



cis−3−Bromo-cis−5−iodo-methyl cyclohexane

(All cis)

(II)

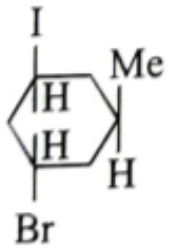


cis−3−Bromo-trans−5− iodo-methyl cyclohexane

(Me, Br ⇒ **cis** ; Me, I ⇒ **trans**)

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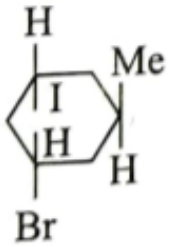
(III)



trans−3−Bromo-cis−5−iodo methyl cyclohexane

$(\mathrm{Me, Br} =$ **trans;** $\mathrm{Me, I} = cis)$

(IV)



trans−3−Bromo-trans−5−iodo-methyl cyclohexane

(Me, Br $\Rightarrow$ **trans;** $\mathrm{Me, I} \Rightarrow$ **trans**)

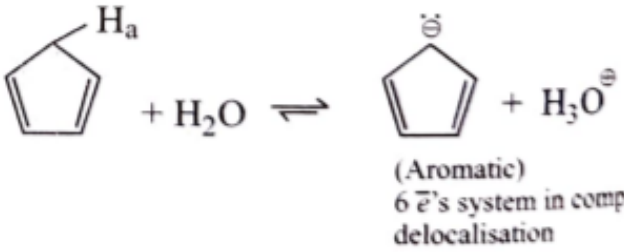
Q.18 Answer:



Solution:

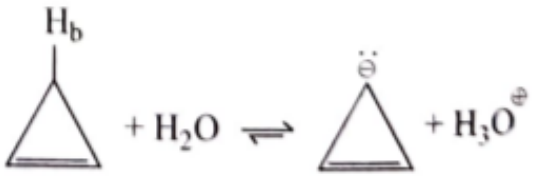
Lower the $\mathrm{p}K_{\mathrm{a}}$**, stronger is the acid** $(\mathrm{H_a}$ **in (I) has lowest) value.**

I.



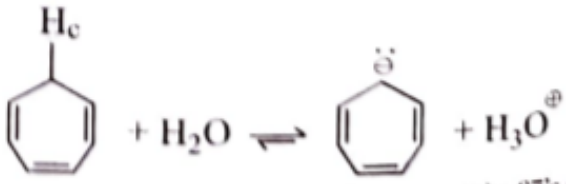
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II.



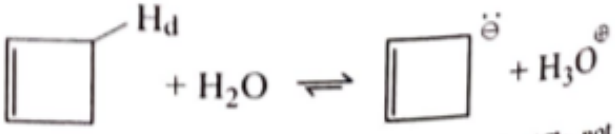
(Anti-aromatic, $4\bar{e}$'s system with complete delocalisation)

III.



(Anti-aromatic, $8\bar{e}$'s system, complete delocalisation)

IV.



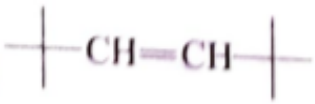
(Non-aromatic, $4\bar{e}$'s, not complete delocalisation)

Stability order: Aromatic > Non-aromatic > Anti-aromatic

pK_a order: $(\mathrm{I}) < (\mathrm{IV}) < (\mathrm{III}) < (\mathrm{II})$ **(** **or** $\mathrm{H_a} < \mathrm{H_d} < \mathrm{H_c} < \mathrm{H_b}$)

H_c in (III) is a stronger acid than $\mathrm{H_b}$ in (II) because its anion is much more stabilised due to more number of resonating structures.

Q.19 Answer:



Solution:

In option (d), as there are no alpha hydrogens, hyperconjugation does not take place.

Q.20 Answer:

free radical

Solution:

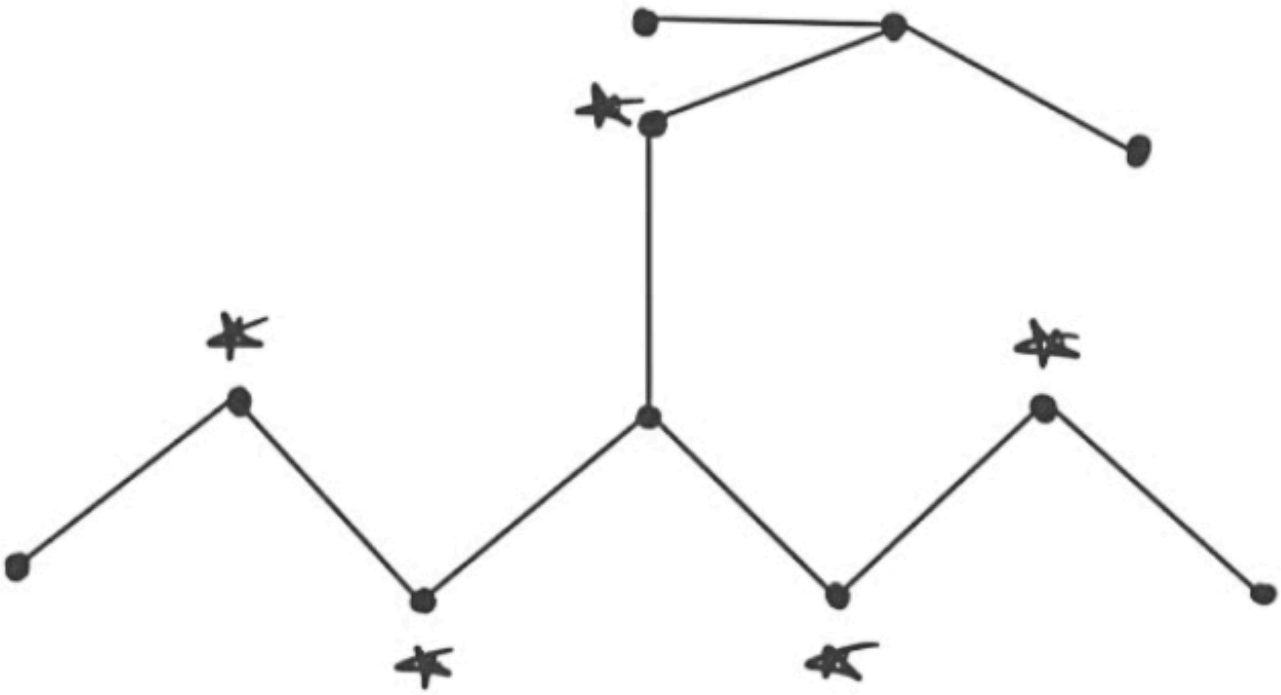
A free radical contains one unpaired electron. Therefore, it is paramagnetic in nature.A carbonium ion and a carbanion contain all paired electrons.Hence, they are diamagnetic in nature.

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Q.21 Answer:

—

Solution:



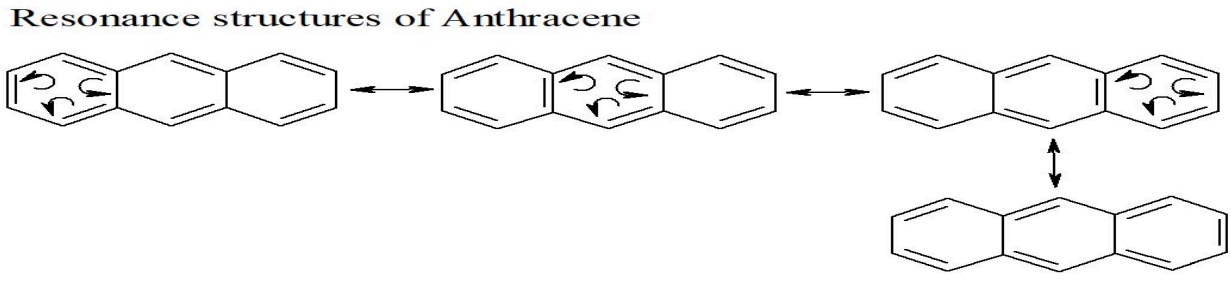
Total 5 secondary carbons are present

Q.22 Answer:

—

Solution:

4 resonance structures are possible for Anthracene.



Q.23 Answer:

—

Solution:

The organic compound is fused with sodium. Carbon and nitrogen present in the organic compound gives sodium cyanide ($NaCN$). This is further used in the test for nitrogen.

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Q.24 Answer:

—

Solution:

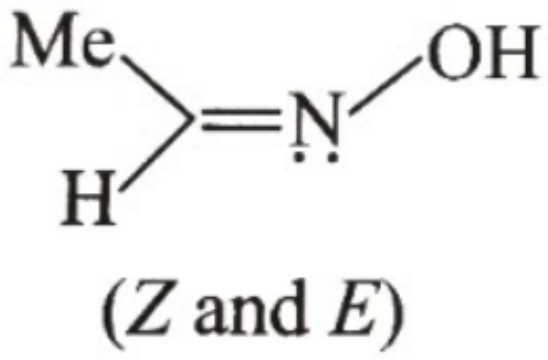
Formic acid, Benzoic acid, Para-nitrophenol, Picric acid, Ortho-nitrophenol are more acidic than phenol while ethanol, water, ortho and para cresol are less acidic than phenol.

Q.25 Answer:

—

Solution:

It will have E-Z isomerism.



(*Z* and *E*)

Q.26 Answer:

—

Solution:

(A) is non-aromatic as it is not planar.
(B) is antiaromatic as it contains conjugated system of $4n\pi$ electrons or 2π electrons.
(C) is aromatic as it contains conjugated system of $4n + 2\pi$ electrons $(2\pi$ electrons). It is cyclic and planar.
(D) is non-aromatic as π electrons are not fully conjugated.
(E) is aromatic as it contains conjugated system of $4n + 2\pi$ electrons $(6\pi$ electrons). It is cyclic and planar.
(F) is anti aromatic as it contains conjugated system of 4 n π electrons (4π electrons). It is cyclic and planar.
(G) is aromatic as it contains conjugated system of $4n + 2\pi$ electrons $(6\pi$ electrons). It is cyclic and planar.
(H) is aromatic as it contains conjugated system of $4n + 2\pi$ electrons $(6\pi$ electrons). It is cyclic and planar. These are π electrons present in one ring. π electrons present in other ring are not counted.
(I) is aromatic as it contains conjugated system of $4n + 2\pi$ electrons (14π electrons). It is cyclic and planar.

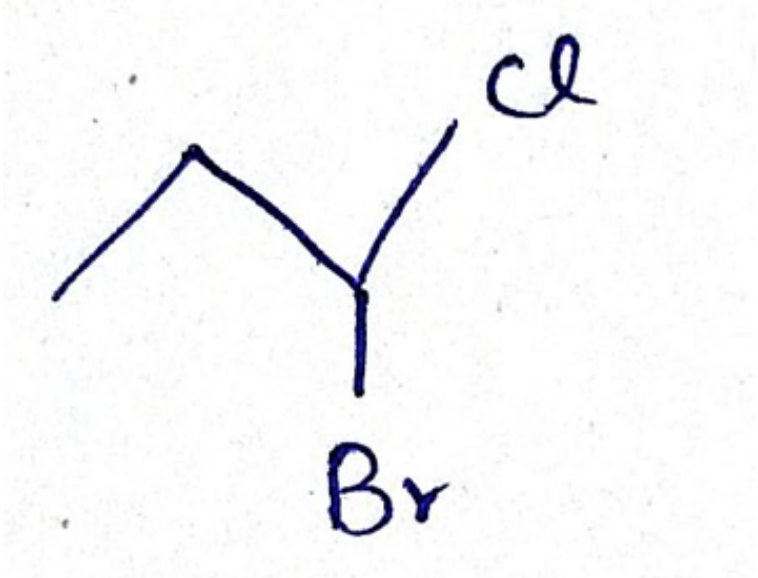
Q.27 Answer:

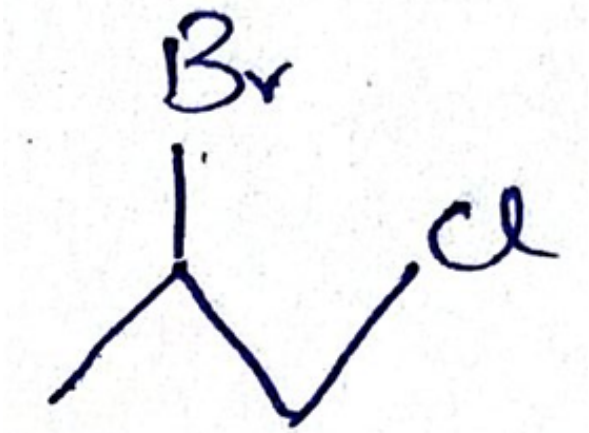
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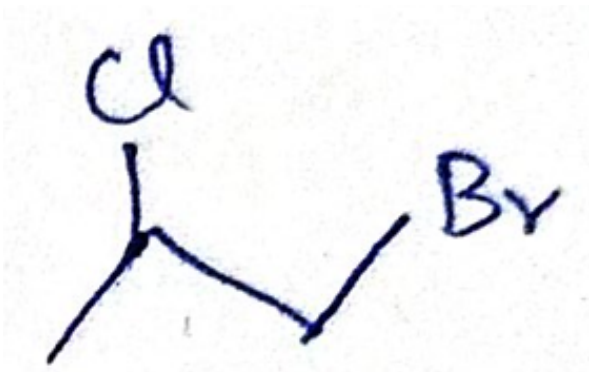
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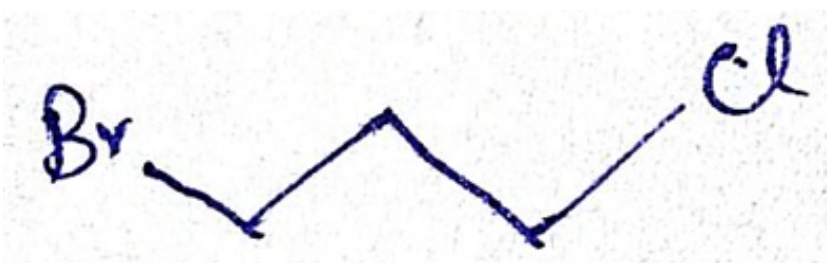
Solution:

C_3H_6BrCl – 5 isomers









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Q.28 Answer:

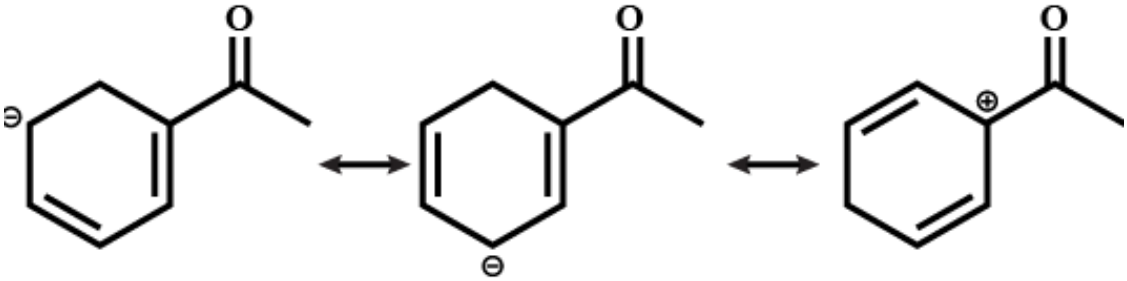
—

Solution:

1) $(CH_3)_3\,C^-$ **is the least stable as it supply more electrons** $+I$ **to the carbanion.**

2) $(CH_3)_2\,HC^-$ **is stable than** $(CH_3)_3\,C^-$ **as it has less** $+I$ **effect compared to the other. But still it is less stable than the others, the reason being the same as above.**

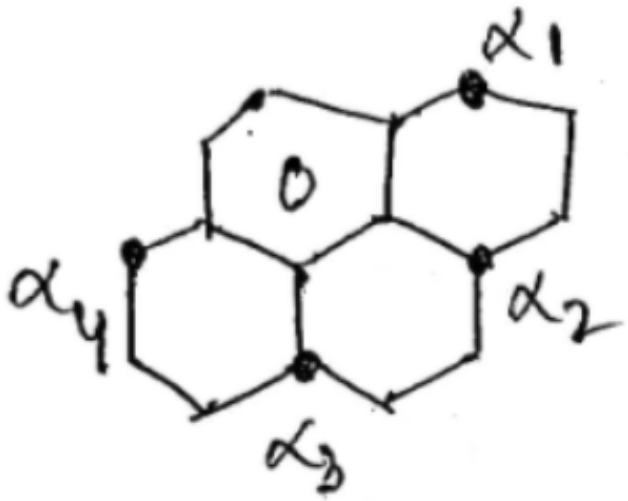
3) $C_6H_5H_2C^-$ **is a resonating structure. Normally resonance effect is more compared to other effects. But since** Cl_3C^- **has** 3 Cl **group** $-I$ **effect, inductive effect overruns resonance effect.**



Q.29 Answer:

—

Solution:



$\rightarrow$ **there are** 4α**-carbon in the compound.**

α_1 **has** $2-$**hydrogen,**

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α_2 **has** $one - H$ **atom**

α_3 **has** $one - H$ **atom.**

α_4 **has** $2 - H$ **atom.**

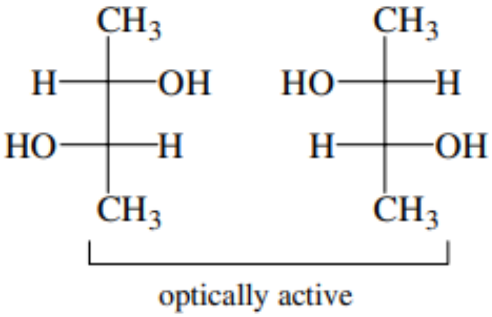
$\therefore$ **Total** $6\ \alpha - H$ **are present.**

Q.30 Answer:

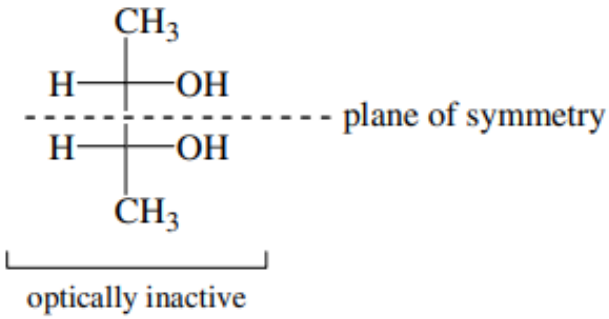
—

Solution:

There are three stereoisomers out of which only two are optically active.



optically active



optically inactive

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